MODULE 1

DESCRIBING DATA

OUTLINE

1. Measuring psychological constructs
2. Distribution shapes for numeric variables
3. Graphical displays for numeric variables
4. Graphical and tabular displays for discrete and categorical variables
5. Study questions

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UVEAL MELANOMA and DEPRESSION

Uveal melanoma, a rare eye cancer, presents potential vision loss and life threat. This prospective, longitudinal study interrogated the predictive utility of visual impairment, as moderated by optimism/pessimism, on depressive symptoms in 299 adults undergoing diagnostic evaluation.



Annette Stanton

James MacDonald

MacDonald, J.J., Jorge-Miller, A., Enders, C.K., McCannel, T., Beran, T., & Stanton, A.L. (2021.. Perceived and objective visual impairment predicting depressive symptoms across one year in uveal melanoma diagnostic biopsy: Optimism and pessimism as moderators. *Health Psychology*, *40*, 408-417.

KEY VARIABLES



Depressive Symptoms

The CES-D is a 20-item inventory that asks people to rate how often they experience depressive symptoms such as restless sleep, poor appetite, and feeling lonely.



Dispositional Optimism

The Life Orientation Test Revised (LOT-R) is a 3-item questionnaire that asks people to rate their tendency to expect positive outcomes in life.

MEASURING DEPRESSIVE SYMPTOMS

- The article's description of the measurement instrument:

The 20-item Center for Epidemiologic Studies Depression Scale (CES-D; Radloff, 1977; $\alpha = .89$. was completed at each assessment point. The CES-D has strong psychometric properties in cancer samples (Hann et al., 1999). Higher scores indicate greater depressive symptoms. An established cutoff score for clinically elevated depressive symptoms is ≥ 16 (Andresen et al., 1994).

CES-D INVENTORY

0 = Rarely or none of the time (less than 1 day)
1 = Some or a little of the time (1-2 days)
2 = Occasionally or a moderate amount of the time (3-4 days)
3 = Most or all of the time (5-7 days)

During the past week:

1. I was bothered by things that usually don't bother me	0	1	2	3
2. I did not feel like eating; my appetite was poor	0	1	2	3
3. I felt that I could not shake off the blues	0	1	2	3
4. I felt that I was just as good as other people	0	1	2	3
5. I had trouble keeping my mind on what I was doing	0	1	2	3
6. I felt depressed	0	1	2	3
7. I felt that everything I did was an effort	0	1	2	3
8. I felt hopeful about the future	0	1	2	3
9. I thought my life had been a failure	0	1	2	3
10. I felt fearful	0	1	2	3
11. My sleep was restless	0	1	2	3
12. I was happy	0	1	2	3
13. I talked less than usual	0	1	2	3
14. I felt lonely	0	1	2	3
15. People were unfriendly	0	1	2	3
16. I enjoyed life	0	1	2	3
17. I had crying spells	0	1	2	3
18. I felt sad	0	1	2	3
19. I felt that people disliked me	0	1	2	3
20. I could not get "going"	0	1	2	3

SUM SCORES

- Researchers routinely assign scores to a psychological construct by adding a participant's responses to questionnaire items measuring the same thing
- Referred to as a sum, scale, or composite score
- Sum scores have better reliability and validity, measure a broader range of attributes, and more closely approximate a numeric (interval or ratio) scale

RECENT METHODOLOGY WORK

- Liu, Y., & Pek, J. (2024). Summed versus estimated factor scores: Considering uncertainties when using observed scores. Psychological Methods, Advance online publication. https://doi.org/dx.doi.org/10.1037/met0000644

- McNeish, D., & Wolf, M. G. (2020). Thinking twice about sum scores. Behavior Research Methods, 52, 2287–2305. https://doi.org/10.3758/s13428-020-01398-0

- Rhemtulla, M., & Savalei, V. (2024). Estimated Factor Scores Are Not True Factor Scores. Multivariate Behavioral Research, Advanced online publication, 1–22. https://doi.org/doi.org/10.1080/00273171.2024.2444943

SUBCLINICAL SUM SCORE

Sum score = 0 + 1 + 1 + 0 + 0 + 2 + 0 + 1 + 0 + 0 + 1 + 1 + 0 + 0 + 1 + 0 + 0 + 0 + 0 + 1 = **9**

During the past week:

1. I was bothered by things that usually don't bother me	**0**	1	2	3	11. My sleep was restless	0	**1**	2	3
2. I did not feel like eating; my appetite was poor	0	**1**	2	3	12. I was happy	0	**1**	2	3
3. I felt that I could not shake off the blues	0	**1**	2	3	13. I talked less than usual	**0**	1	2	3
4. I felt that I was just as good as other people	**0**	1	2	3	14. I felt lonely	**0**	1	2	3
5. I had trouble keeping my mind on what I was doing	**0**	1	2	3	15. People were unfriendly	0	**1**	2	3
6. I felt depressed	0	1	**2**	3	16. I enjoyed life	**0**	1	2	3
7. I felt that everything I did was an effort	**0**	1	2	3	17. I had crying spells	**0**	1	2	3
8. I felt hopeful about the future	0	**1**	2	3	18. I felt sad	**0**	1	2	3
9. I thought my life had been a failure	**0**	1	2	3	19. I felt that people disliked me	**0**	1	2	3
10. I felt fearful	**0**	1	2	3	20. I could not get "going"	0	**1**	2	3

CLINICALLY ELEVATED SUM SCORE

Sum score = 1 + 2 + 2 + 1 + 1 + 3 + 1 + 2 + 1 + 1 + 2 + 2 + 1 + 1 + 2 + 1 + 1 + 1 + 1 + 2 = **29**

During the past week:

1. I was bothered by things that usually don't bother me	0	**1**	2	3	11. My sleep was restless	0	1	**2**	3
2. I did not feel like eating; my appetite was poor	0	1	**2**	3	12. I was happy	0	1	**2**	3
3. I felt that I could not shake off the blues	0	1	**2**	3	13. I talked less than usual	0	**1**	2	3
4. I felt that I was just as good as other people	0	**1**	2	3	14. I felt lonely	0	**1**	2	3
5. I had trouble keeping my mind on what I was doing	0	**1**	2	3	15. People were unfriendly	0	1	**2**	3
6. I felt depressed	0	1	2	**3**	16. I enjoyed life	0	**1**	2	3
7. I felt that everything I did was an effort	0	**1**	2	3	17. I had crying spells	0	**1**	2	3
8. I felt hopeful about the future	0	1	**2**	3	18. I felt sad	0	**1**	2	3
9. I thought my life had been a failure	0	**1**	2	3	19. I felt that people disliked me	0	**1**	2	3
10. I felt fearful	0	**1**	2	3	20. I could not get "going"	0	1	**2**	3

MEASURING DISPOSITIONAL OPTIMISM

- The article's description of the optimism instrument:

Dispositional optimism and pessimism were assessed prior to diagnosis with the Life Orientation Test—Revised (LOT-R; Scheier et al., 1994; $\alpha = .82$). The LOT-R consists of three optimism items, three pessimism items, and four fillers. Five-point Likert scale items range from I agree a lot to I disagree a lot. Summed scores for optimism ($\alpha = .74$) and pessimism ($\alpha = .80$) were calculated. Each scale's potential range is 0–12; higher scores represent higher levels of the attribute.

LOT-R OPTIMISM QUESTIONNAIRE

4 = I agree a lot

3 = I agree a little

2 = I neither agree nor disagree

1 = I disagree a little

0 = I disagree a lot

In uncertain times, I usually expect the best.	0	1	2	3	4
I'm always optimistic about my future.	0	1	2	3	4
Overall, I expect more good things to happen to me than bad	0	1	2	3	4

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DISTRIBUTIONS

- A distribution describes the concentration of score values
- Distributions can be summarized with graphs, tables, or mathematical functions
- The type of summary depends on whether a variable is numeric (interval or ratio) or categorical (nominal or ordinal)

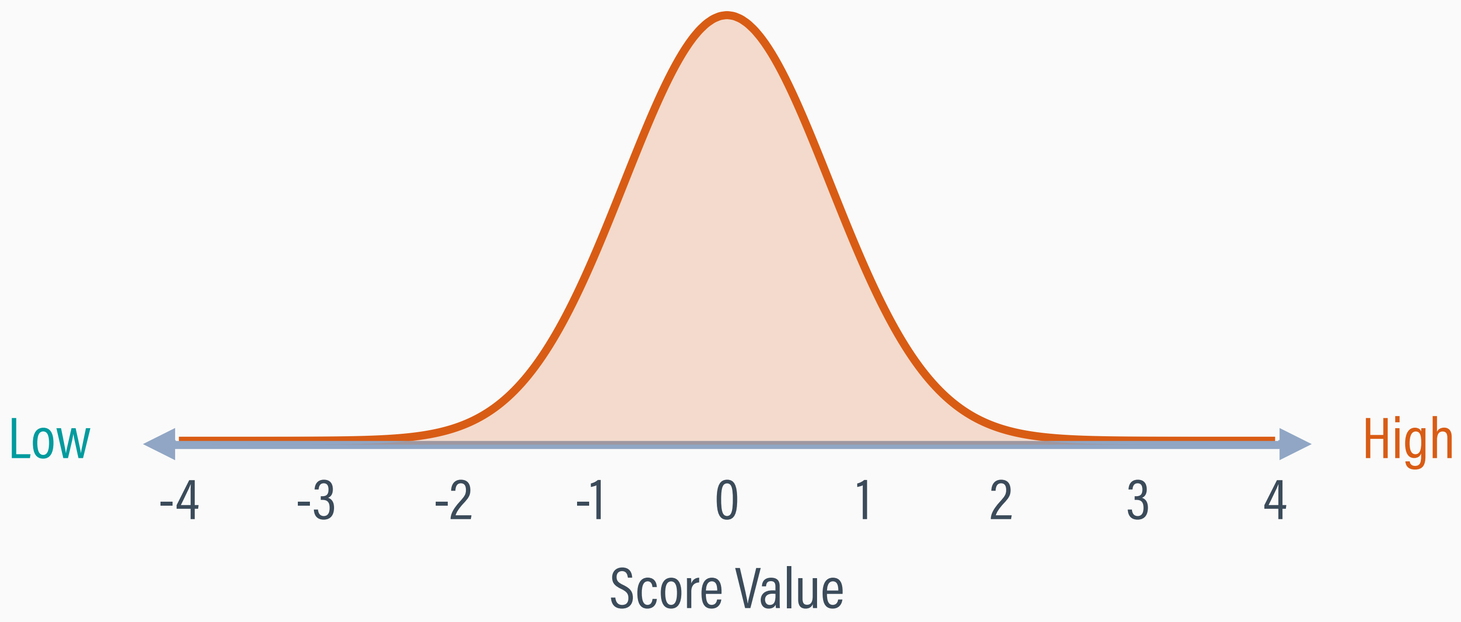
NUMERIC (CONTINUOUS) VARIABLES

1-point increase have the same meaning!

- Interval and ratio scales refer to numeric (continuous) variables that approximate a number line (e.g., response time, age, GRE)
- Any two adjacent score values reflect the same amount of the variable (e.g., the difference between CESD = 5 and 6 is the same amount of depression as CESD = 20 and 21)
- Many behavioral, emotional, and physiological variables in psychology approximate interval-level data (but not perfectly)

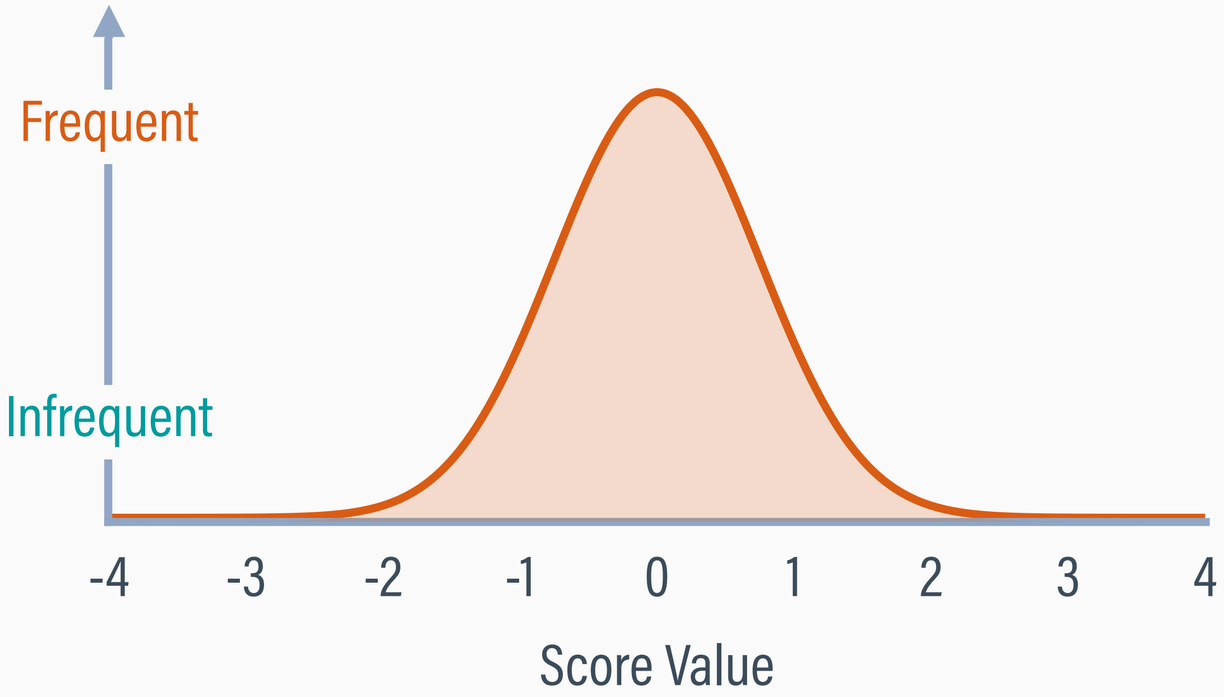
HORIZONTAL AXIS (SCORE RANGE)

- Scores follow a number line and range from low to high along the horizontal axis



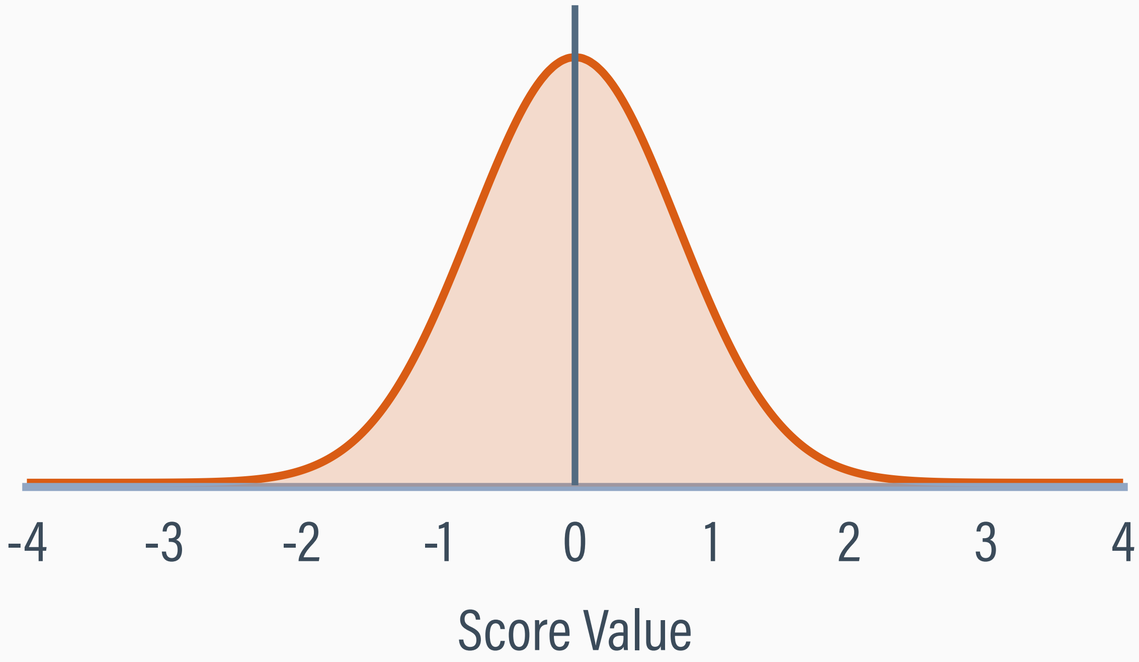
VERTICAL AXIS (FREQUENCY)

- The height of the vertical axis — called frequency or count or density — describes how often each score occurs



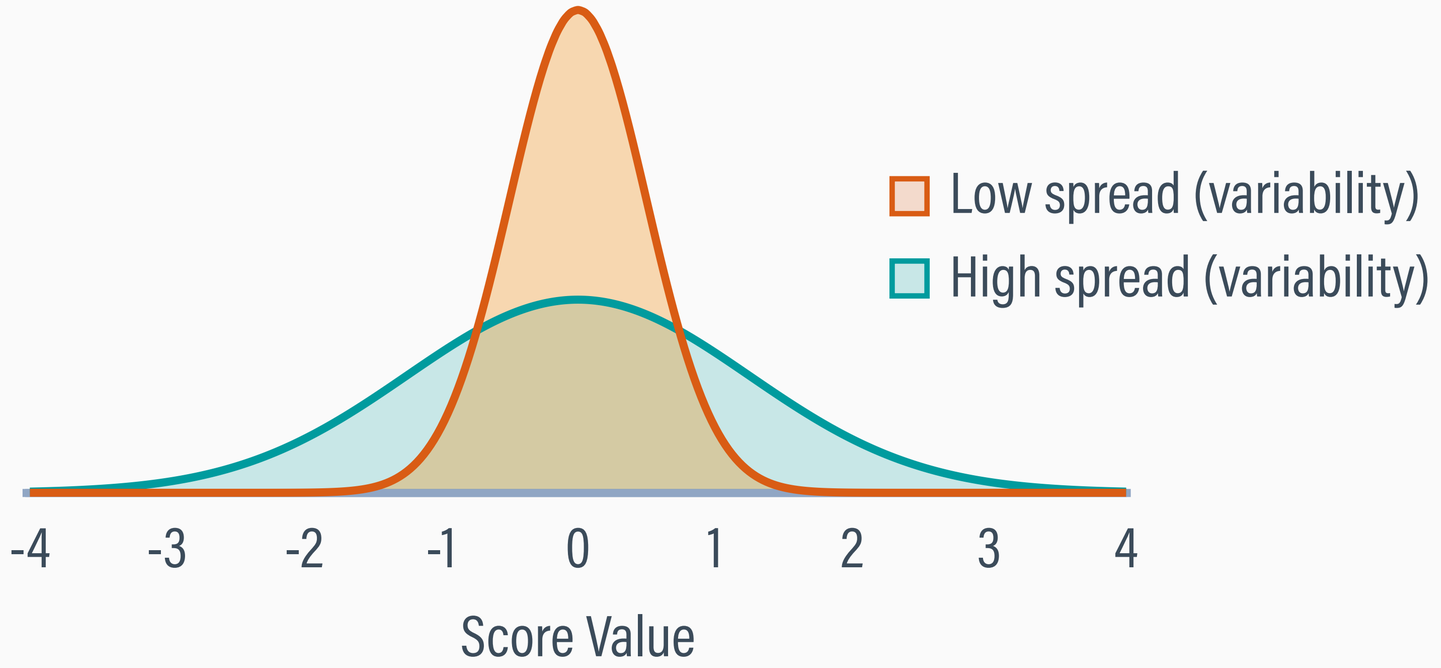
DISTRIBUTION CENTER

- A distribution's center (e.g., mean) is located at a value where scores are highly concentrated



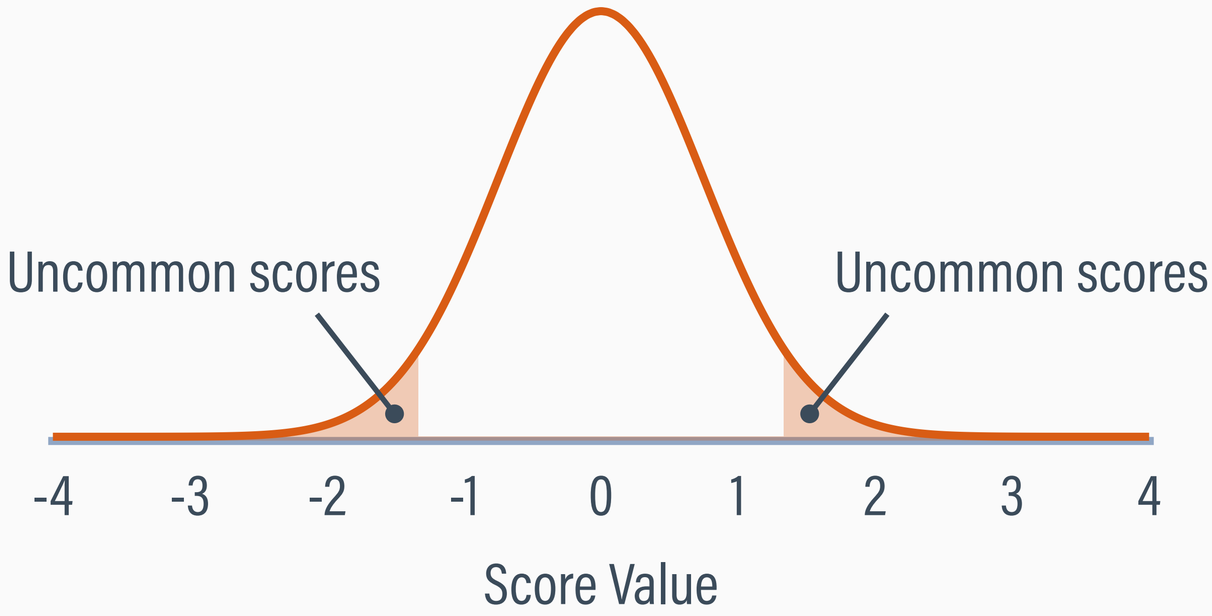
VARIABILITY OR SPREAD

- High variability (spread) indicates greater score differences, low spread indicates similar scores



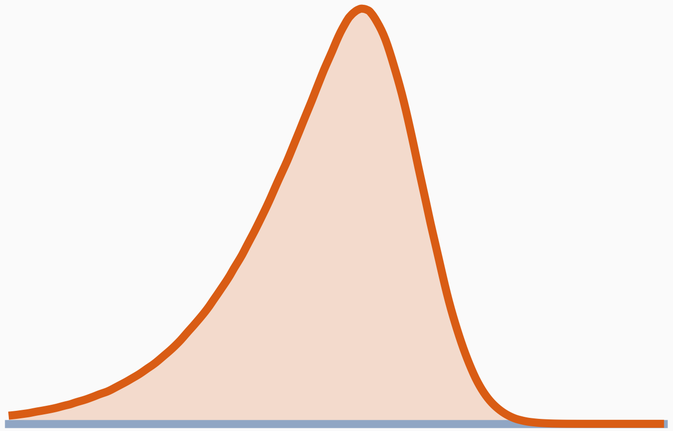
DISTRIBUTION TAILS

- A distribution's tails contain unusual or infrequent score values (outliers)

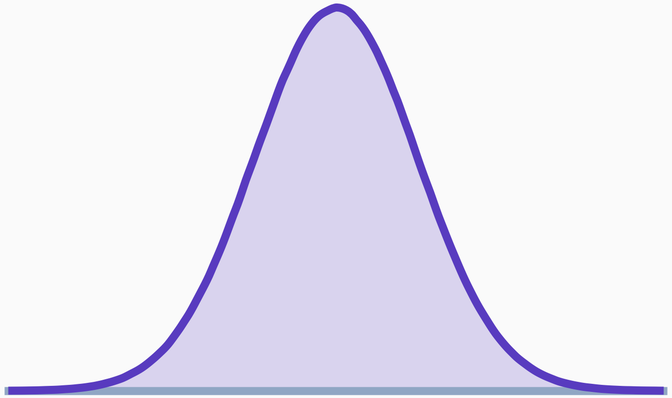


DISTRIBUTION SHAPE

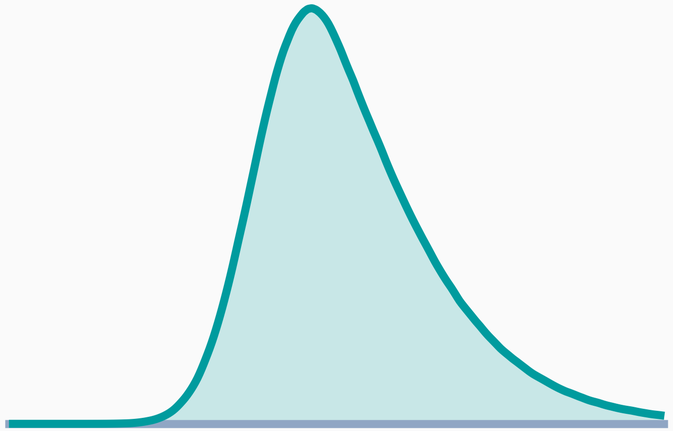
- Asymmetric distributions can be characterized as positively or negatively skewed



Negatively Skewed



Symmetric



Positively Skewed

SKEWNESS AND KURTOSIS

- Skewness is a numerical measure of asymmetry in a distribution (a normal curve has skewness = 0)
- Kurtosis is often described as peakedness but, more accurately, it reflects the proportion of extreme outliers in the tails relative to a normal curve
- Both measures are standardized, such that their numeric values don't depend on a variable's scale

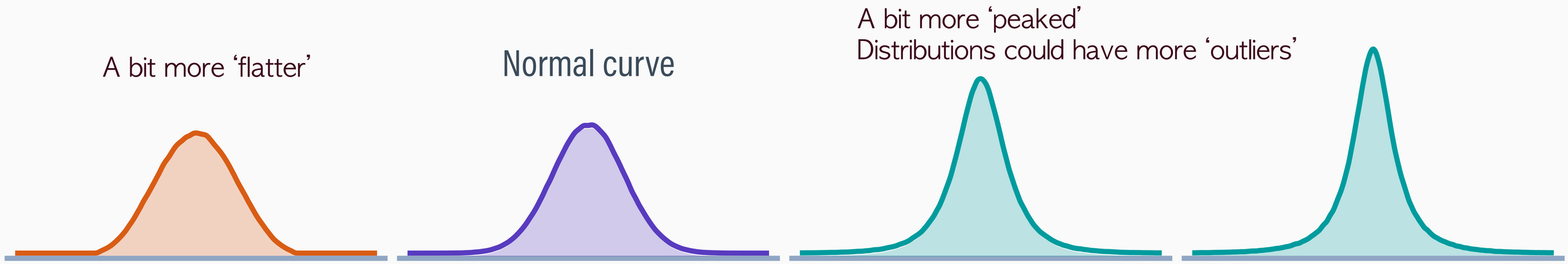
TWO MEASURES OF KURTOSIS

- The kurtosis value for a normal curve is 3
- Researchers often use "excess" kurtosis where the value for a normal curve is 0 (i.e., kurtosis – 3)
- Software packages are not uniform, always check which value is getting reported (we will use the psych package, which reports excess kurtosis)

Which exhibits 'zero' for the bell-curve kurtosis, not three!!!

EXCESS KURTOSIS

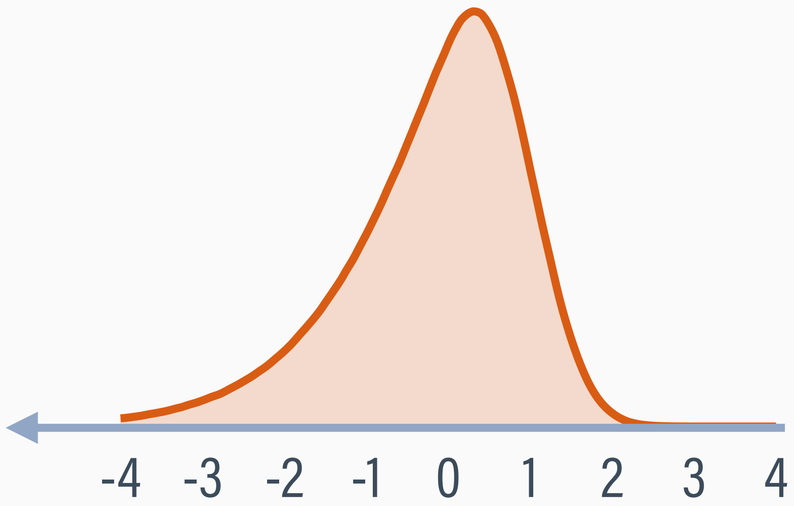
- As kurtosis increases, more scores migrate from the center of the distribution to the tails (thus increasing peakedness)



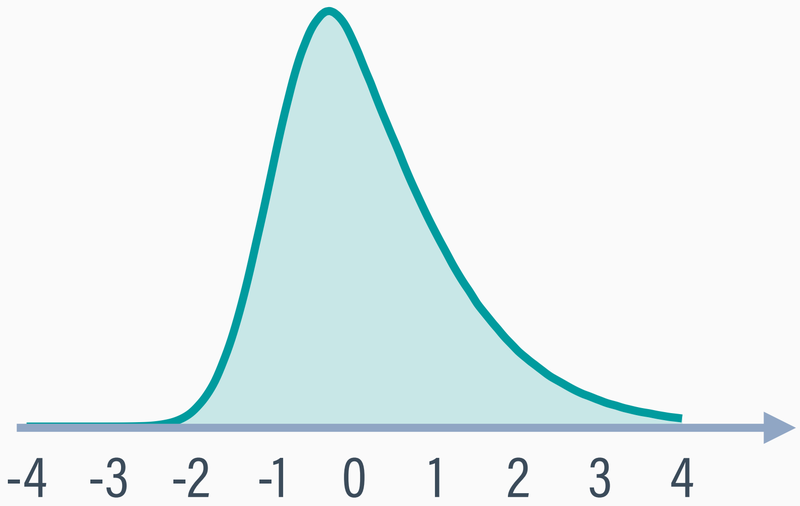
Skewness = 0	Skewness = 0	Skewness = 0	Skewness = 0
Excess Kurtosis = –0.50	Excess Kurtosis = 0	Excess Kurtosis = 4	Excess Kurtosis = 7

POSITIVE VERSUS NEGATIVE SKEW

- A positively skew has a long tail pointing toward the positive side of the number line, and negative skew has the opposite



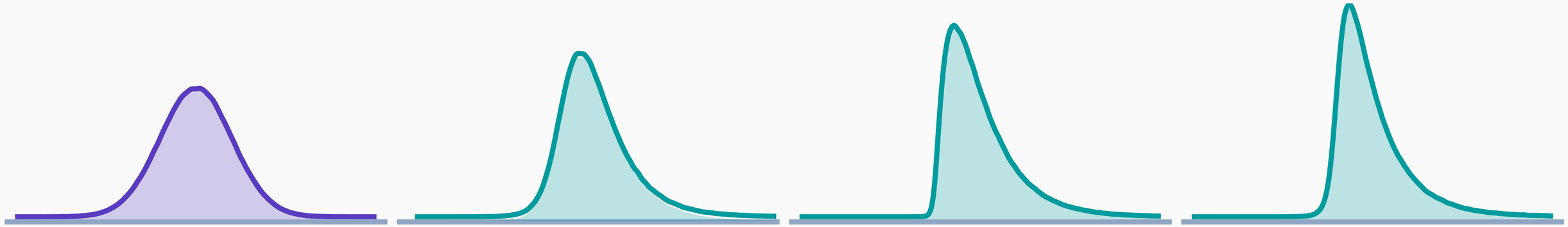
Negatively Skewed



Positively Skewed

POSITIVE SKEWNESS

- As skewness increases, the distribution becomes increasingly asymmetric (kurtosis must increase too)



Skewness = 0
Excess Kurtosis = 0

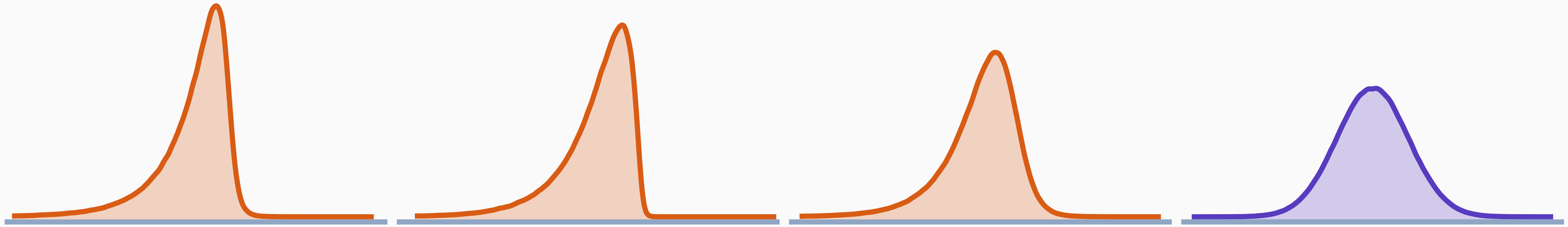
Skewness = 1.25
Excess Kurtosis = 3.50

Skewness = 1.75
Excess Kurtosis = 5.00

Skewness = 2.00
Excess Kurtosis = 7.00

NEGATIVE SKEWNESS

- As skewness increases, the distribution becomes increasingly asymmetric (kurtosis must increase too)



Skewness = –2.00
Excess Kurtosis = 7.00

Skewness = –1.75
Excess Kurtosis = 5.00

Skewness = –1.25
Excess Kurtosis = 3.50

Skewness = 0
Excess Kurtosis = 0

WHY DOES SHAPE MATTER?

- Many statistical procedures (e.g., t-tests, ANOVA, regression) assume that the dependent variable is normally distributed
- Nonnormality often does not affect the accuracy of the estimated statistics themselves
- It can affect the accuracy of significance tests (depending on shape, an effect may be more or less likely to be detected)

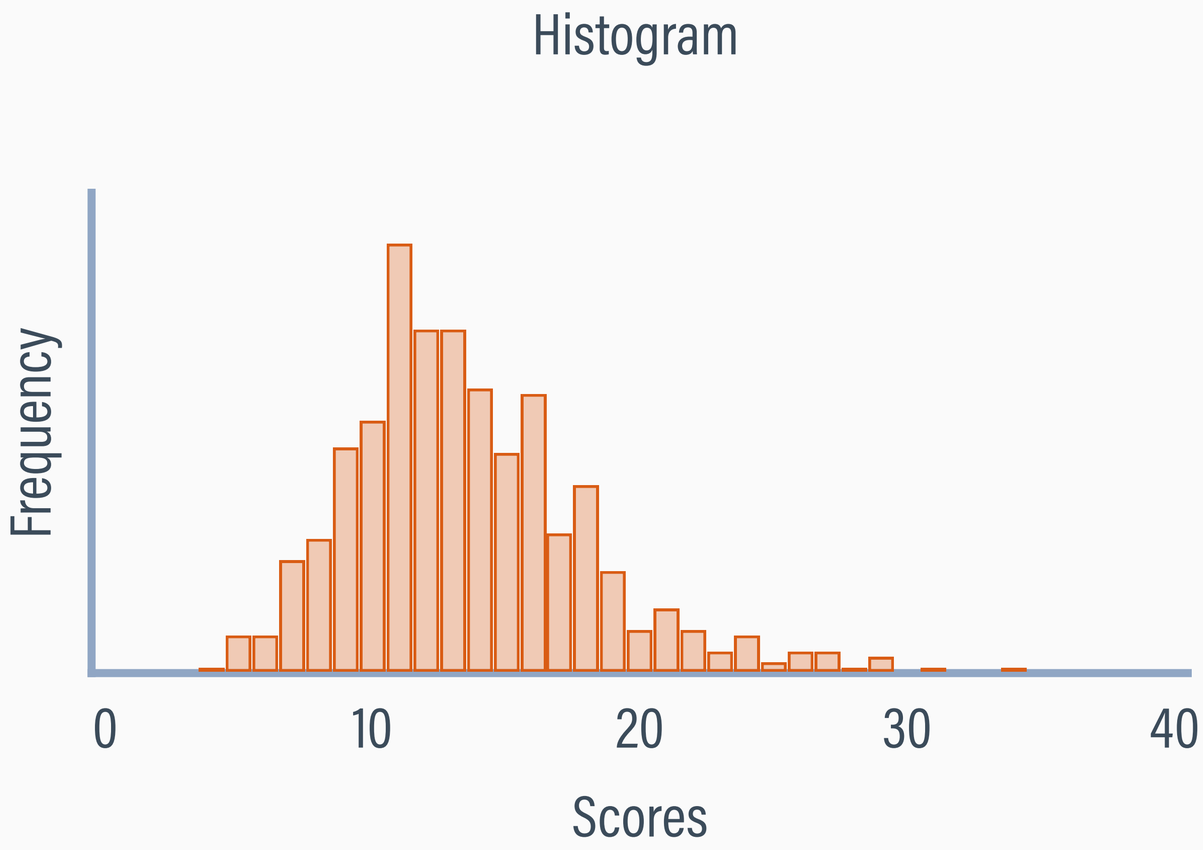
OUTLINE

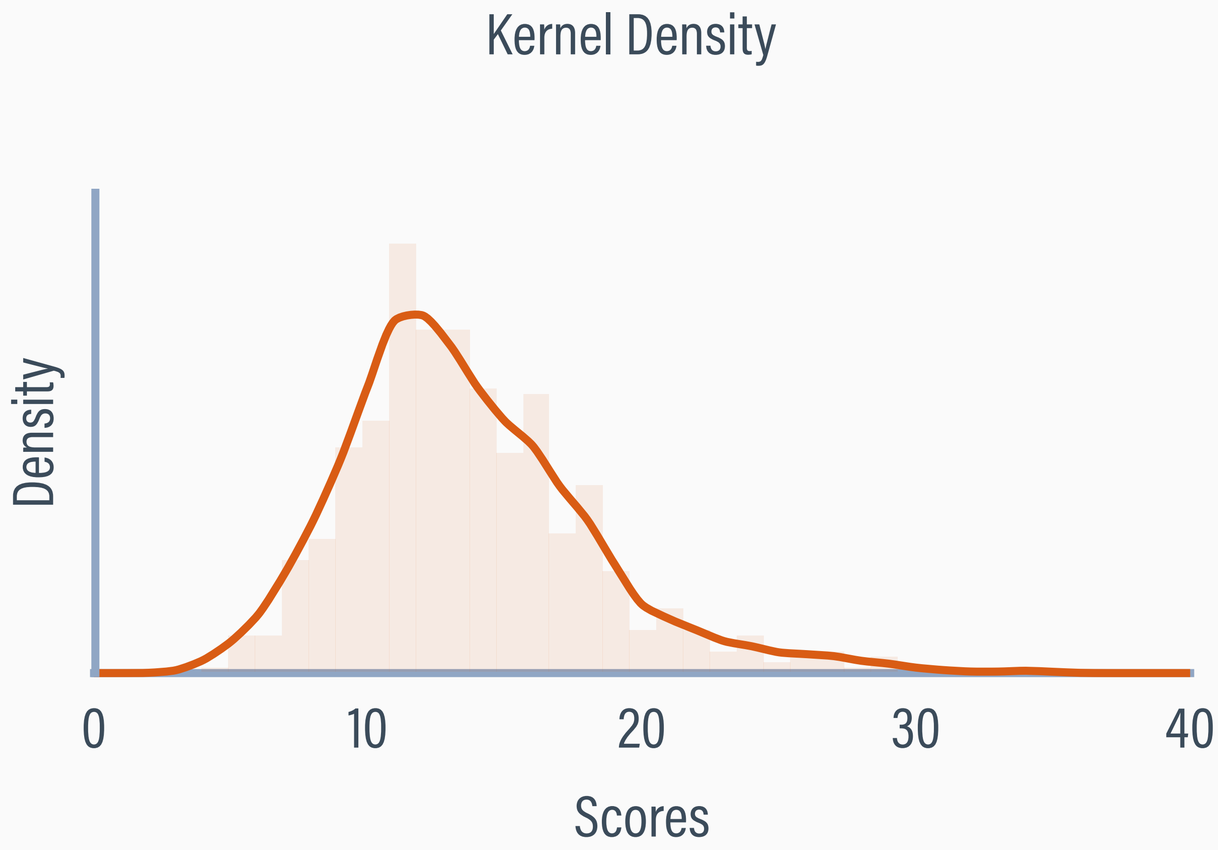
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HISTOGRAMS AND KERNEL DENSITY PLOTS

- A histogram is a bar plot for numeric variables (no breaks or gaps along the horizontal axis, which is a number line)
- A kernel density plot applies a smoothing algorithm that connects the histogram bars with a continuous curve
- Frequency = density = count = vertical elevation

HISTOGRAM AND KERNEL DENSITY GRAPHS





UVEAL MELANOMA and DEPRESSION

Uveal melanoma, a rare eye cancer, presents potential vision loss and life threat. This prospective, longitudinal study interrogated the predictive utility of visual impairment, as moderated by optimism/pessimism, on depressive symptoms in 299 adults undergoing diagnostic evaluation.



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KEY NUMERIC VARIABLES



Depressive Symptoms

The CES-D is a 20-item inventory that asks people to rate how often they experience depressive symptoms such as restless sleep, poor appetite, and feeling lonely.



Dispositional Optimism

The Life Orientation Test Revised (LOT-R) is a 3-item questionnaire that asks people to rate their tendency to expect positive outcomes in life.

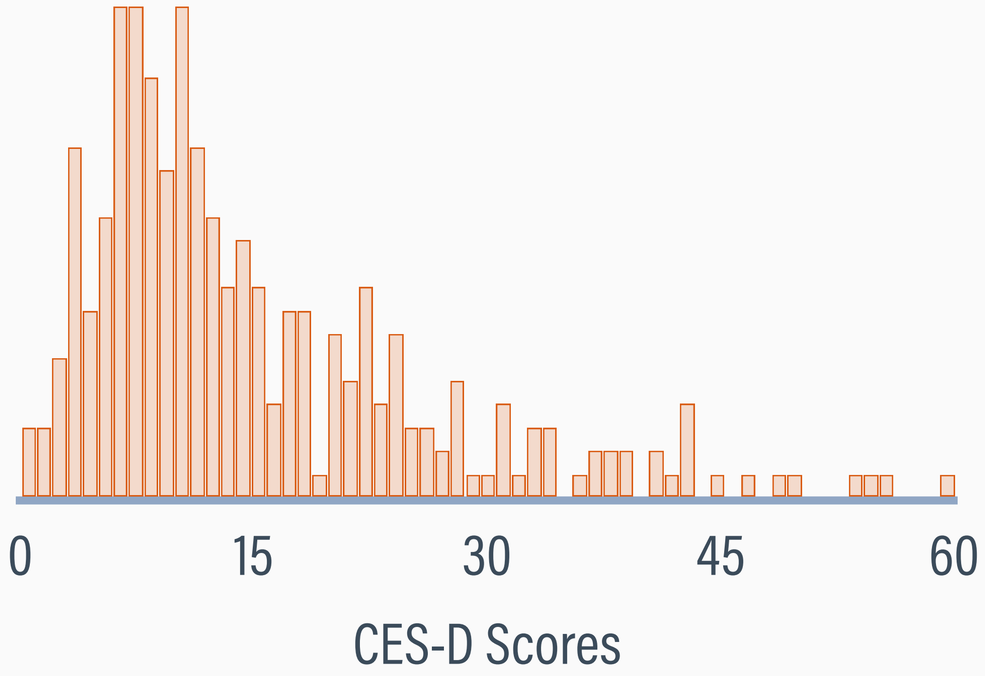


Visual Impairment

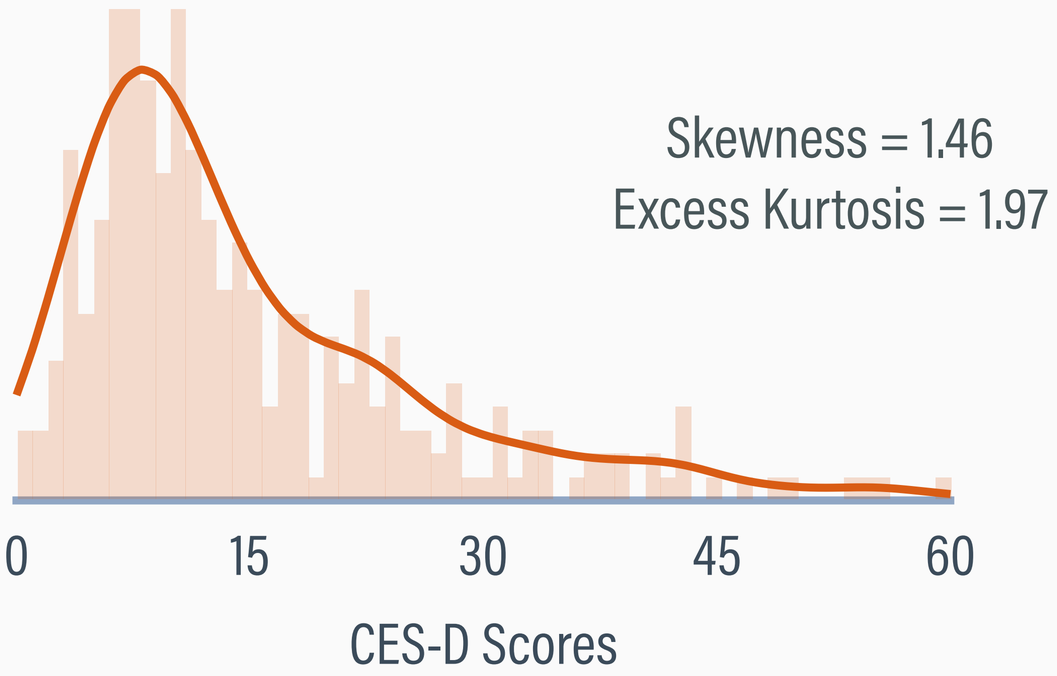
The Measure of Outcome in Ocular Disease vision subscale (MOOD-V) is a 13-item measure. Scores were summed and transformed to a 0–100 scale.

DEPRESSIVE SYMPTOMS DISTRIBUTION

Histogram



Kernel Density



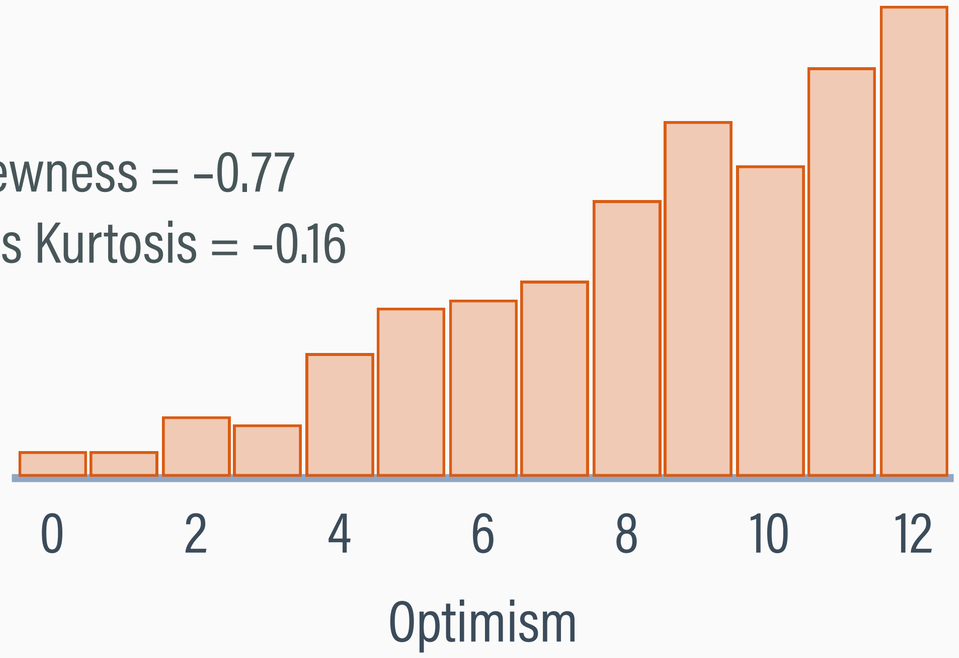


In small groups of two or three, discuss the skewness and kurtosis statistics for the depression variable. Interpret each statistic in practical terms (e.g., what it says about tail heaviness or asymmetry). How do these statistics align with your visual impression from the plot? Based on the distribution shape, how would you characterize the overall magnitude of depressive symptoms in this sample?

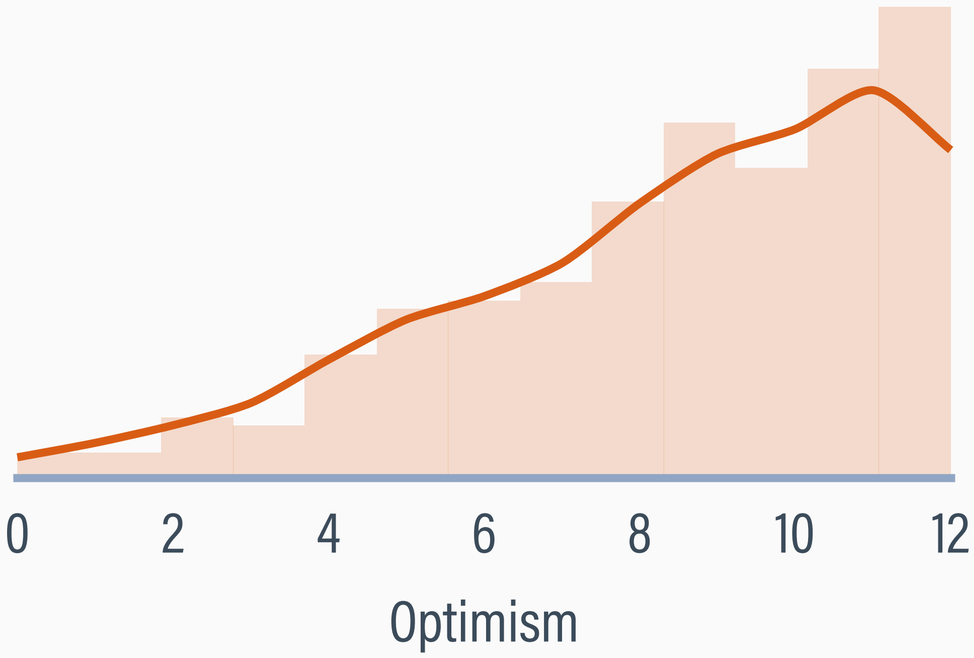
OPTIMISM DISTRIBUTION

Histogram

Skewness = −0.77
Excess Kurtosis = −0.16



Kernel Density

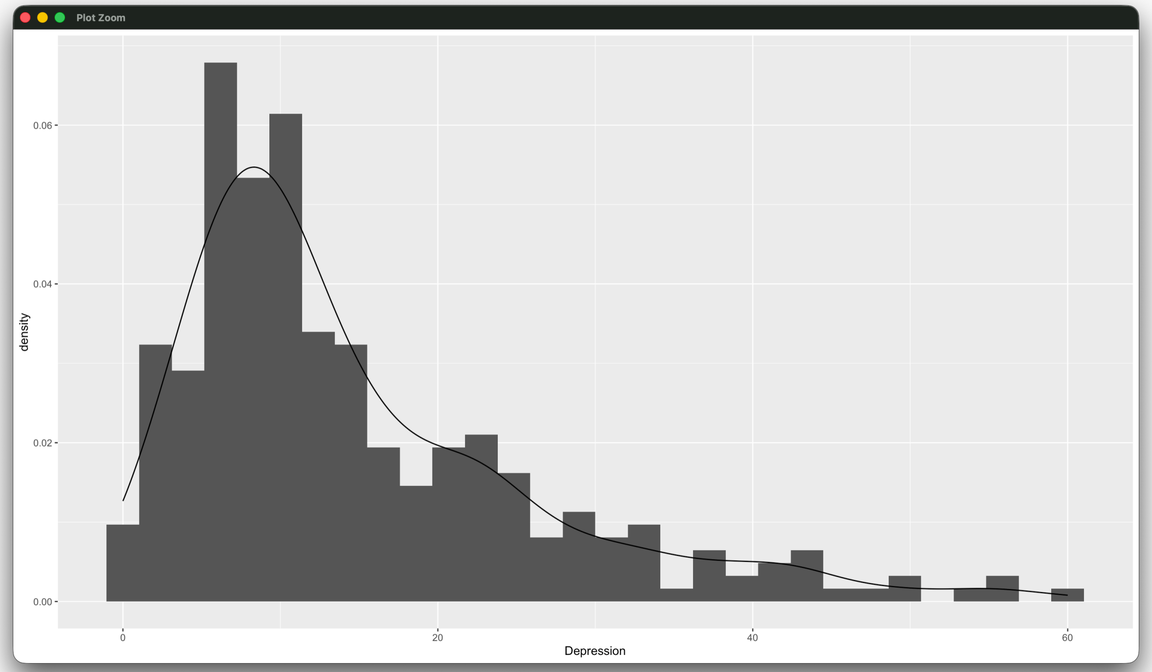


R OUTPUT

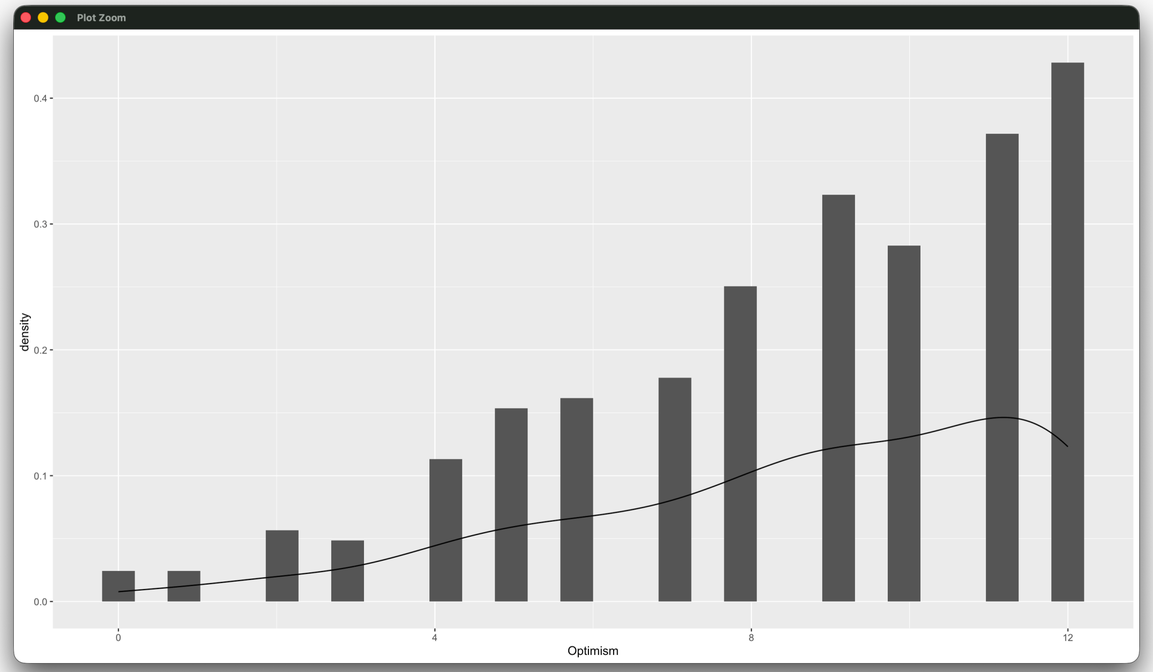
	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
Participant	1	299	150.00	86.46	150.00	150.00	111.19	1.00	299.00	298.00	0.00	-1.21	5.00
Diagnosis*	2	299	1.64	0.48	2.00	1.68	0.00	1.00	2.00	1.00	-0.59	-1.66	0.03
Age	3	299	58.99	13.94	60.00	59.47	11.86	19.00	91.00	72.00	-0.33	-0.15	0.81
Gender*	4	299	1.54	0.50	2.00	1.55	0.00	1.00	2.00	1.00	-0.15	-1.98	0.03
Comorbids	5	299	0.94	0.74	1.00	0.93	1.48	0.00	2.00	2.00	0.10	-1.17	0.04
Optimism	6	299	8.65	3.06	9.00	8.87	2.97	0.00	14.00	14.00	-0.62	-0.23	0.18
Depression	7	299	14.85	11.44	11.00	13.12	7.41	0.00	60.00	60.00	1.46	1.97	0.66
VisImpair	8	299	5.09	2.23	4.85	4.98	2.21	0.36	12.18	11.82	0.43	-0.03	0.13

R OUTPUT

Depression



Optimism



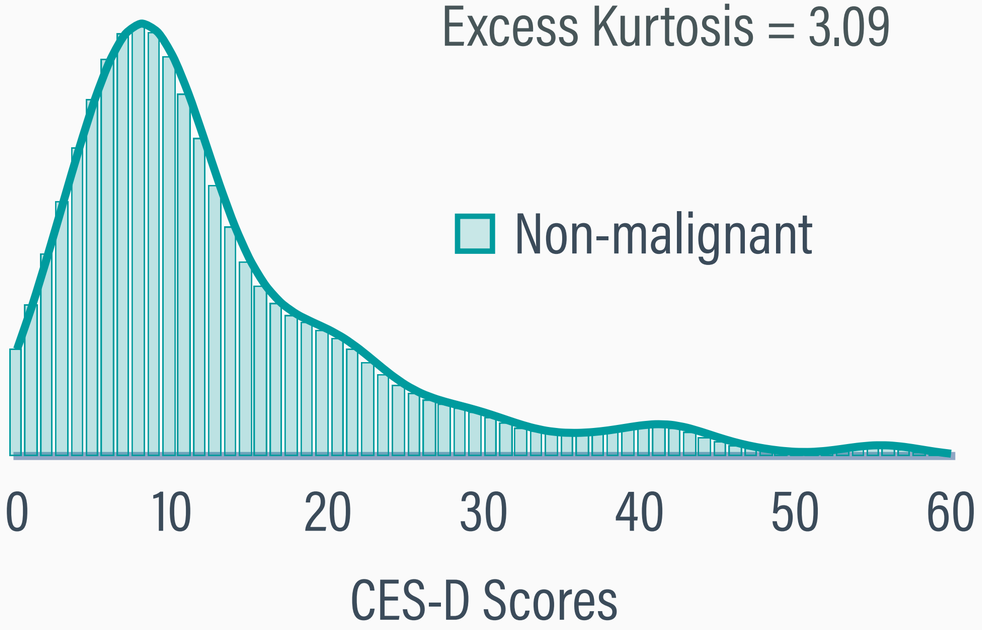
COMPARATIVE RESEARCH QUESTIONS

- Comparative research questions ask whether two or more groups (or occasions) differ from one another
- Question: Do participants diagnosed with cancer differ in depressive symptoms compared to those receiving a negative (no cancer) diagnosis?
- We can begin to answer this question by comparing the distributions of the two groups

GROUP DISTRIBUTIONS

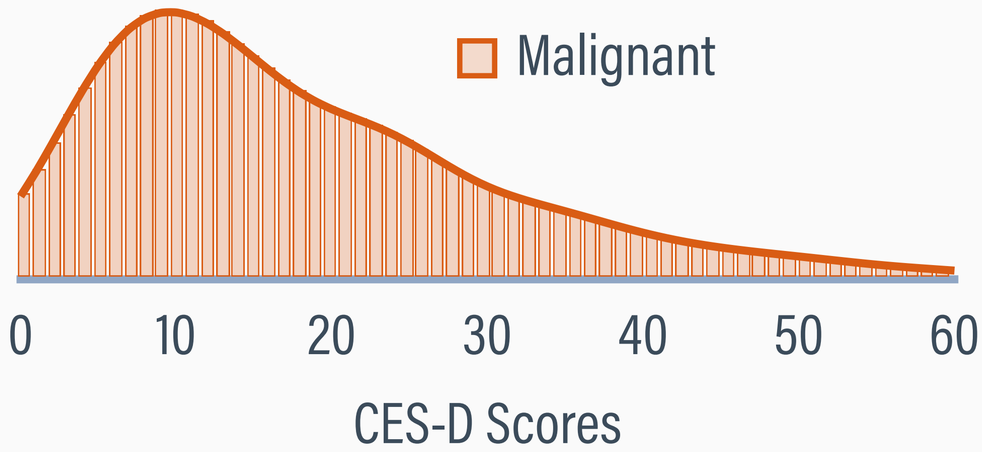
relationship btw skewness, kurtosis, and stdev?
stdev means the deviation between 'mean value'
but at the first graph, the outliers may make much more deviation
which, at a glance, seems to have smaller stdev

Skewness = 1.70
Excess Kurtosis = 3.09

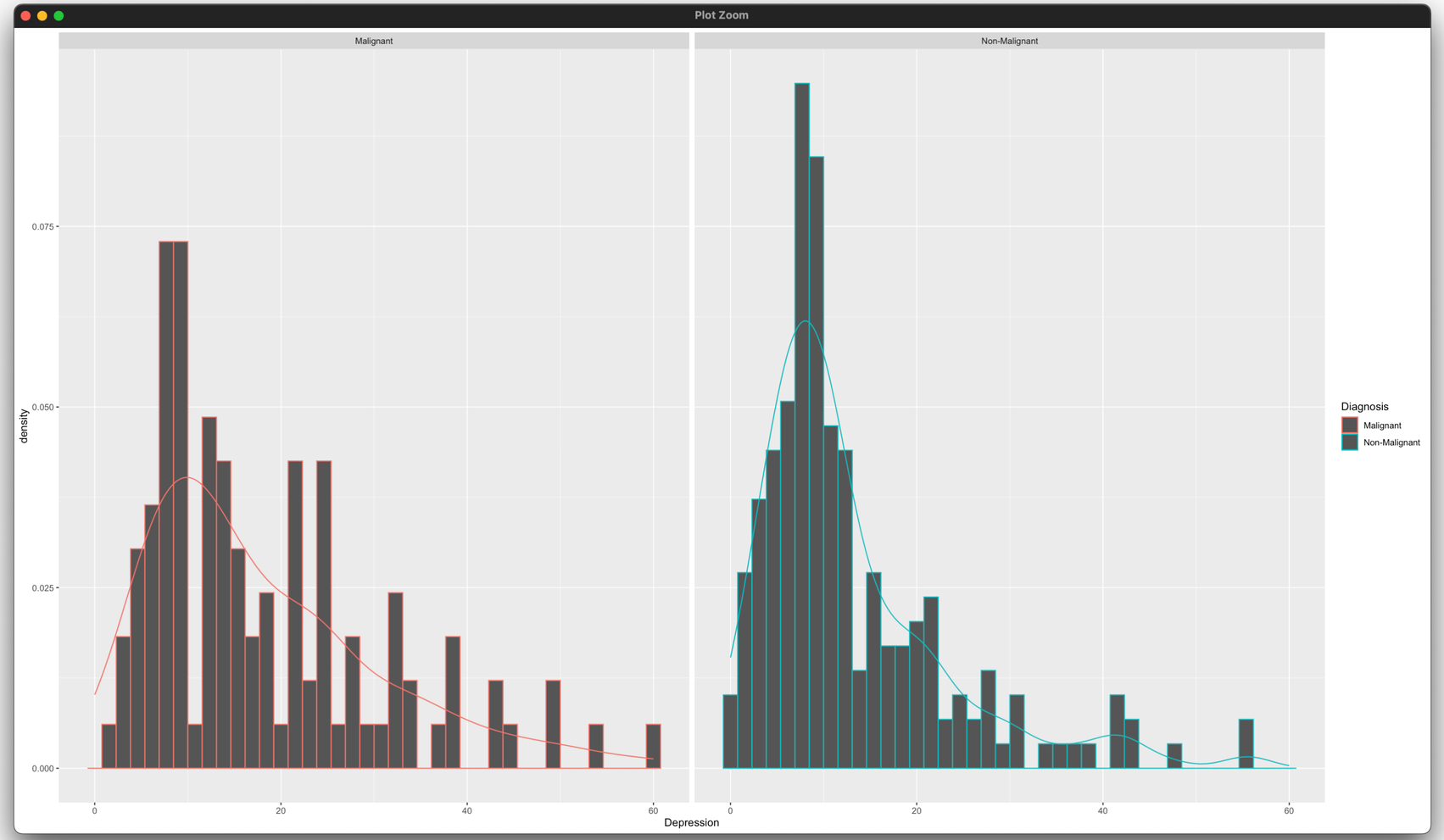


Skewness = 1.12
Excess Kurtosis = 0.80

Not much asymmetric compare to left one



R OUTPUT


Plot Zoom
Malignant
Non-Malignant
density
0.075
0.050
0.025
0.000
0
20
40
60
Depression
Diagnosis
Malignant
Non-Malignant



One of the main goals of the research study is to determine whether the diagnostic groups differ. In small groups of two or three, examine and discuss the two distributions. Identify specific features of the distributions (e.g., location, spread, shape) that suggest differences between groups, as well as features that suggest similarities. Conclude with a brief statement about whether the visual evidence alone suggests that a cancer diagnosis affects depressive symptoms.

OUTLINE

1. Measuring psychological constructs
2. Distribution shapes for numeric variables
3. Graphical displays for numeric variables
4. **Graphical and tabular displays for discrete and categorical variables**
5. Study questions

CATEGORICAL VARIABLES

- Nominal variables group features into mutually exclusive, unordered categories with no inherent rank order (e.g., gender identity, racial ethnic identity, clinical diagnosis.

- Ordinal variables group features into ordered categories that can be ranked from low to high (e.g., five-point Likert scales, educational attainment groupings)

BAR PLOTS AND FREQUENCY DISTRIBUTIONS

- A bar plot is a graphical display for categorical variables that shows the number (or percent) of responses in each bin
- A frequency distribution shows the same information (the range of scores, number and % in each bin) in a table

KEY VARIABLES



Cancer Diagnosis

Based on a diagnostic clinical evaluation for a possible intraocular malignancy, participants were classified as having malignant or nonmalignant diagnoses.



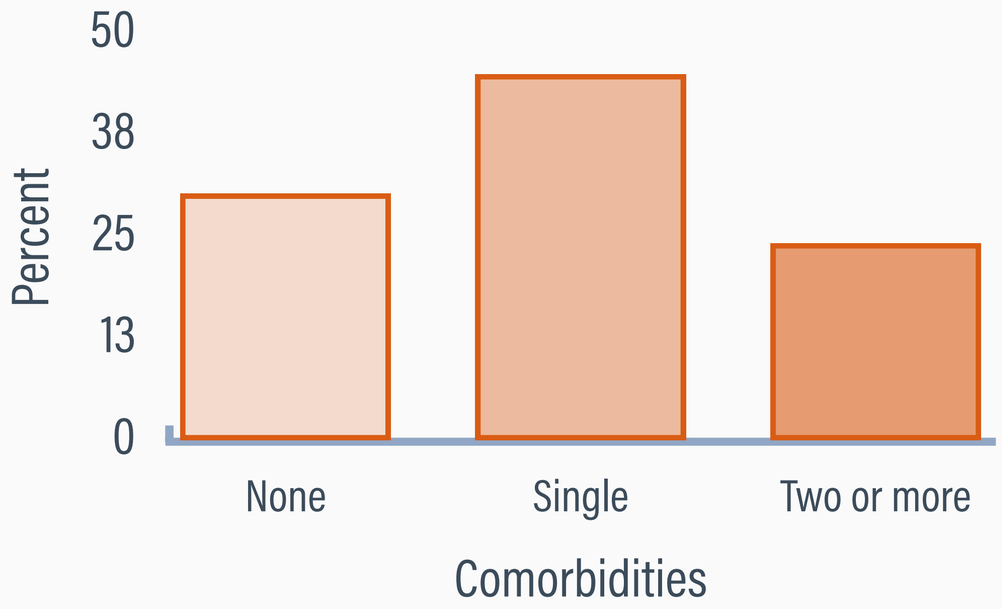
Comorbidities

Participants reported their number of medical comorbidities. Comorbidities were coded 0 (no comorbidities), 1 (a single comorbidity), or 2 (two or more comorbidities).

BAR GRAPH VS. FREQUENCY DISTRIBUTION

- Frequency distributions provide the same information in tabular format

Bar Graph



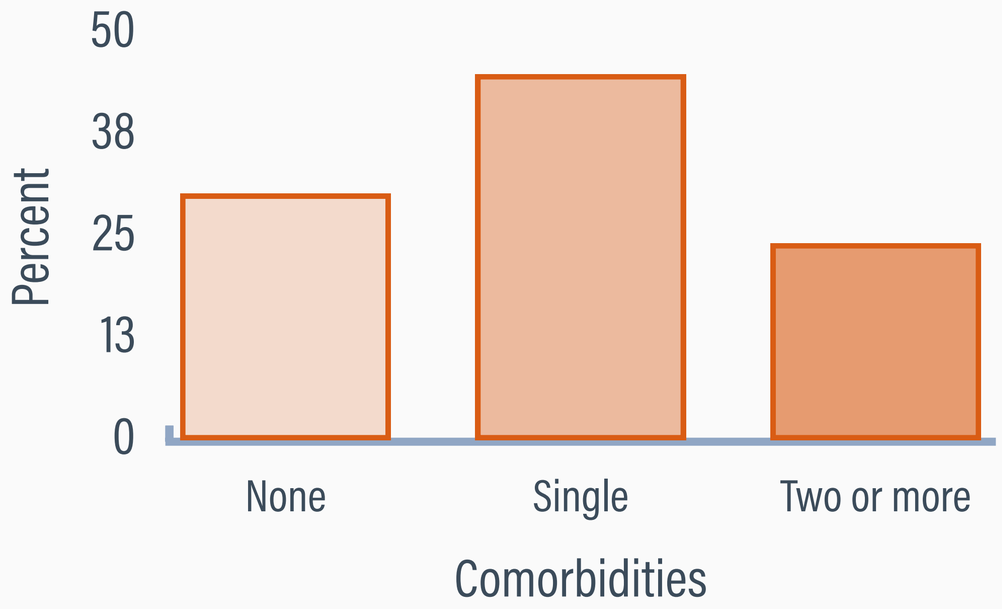
Frequency Distribution

Score	Count	%	Cum. %
0 = None	91	30.43	30.43
1 = Single comorbidity	135	45.15	75.59
2 = Two or more	73	24.41	100.0

BAR GRAPH VS. FREQUENCY DISTRIBUTION

- Frequency distributions provide the same information in tabular format

Bar Graph



Frequency Distribution

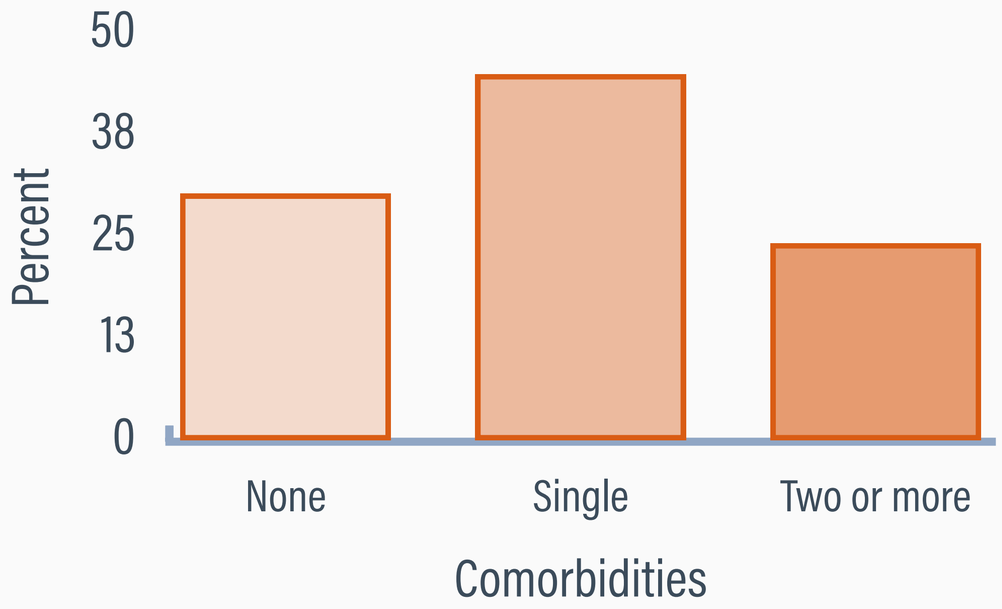
Score	Count	%	Cum. %
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Horizontal axis values

BAR GRAPH VS. FREQUENCY DISTRIBUTION

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Bar Graph



Frequency Distribution

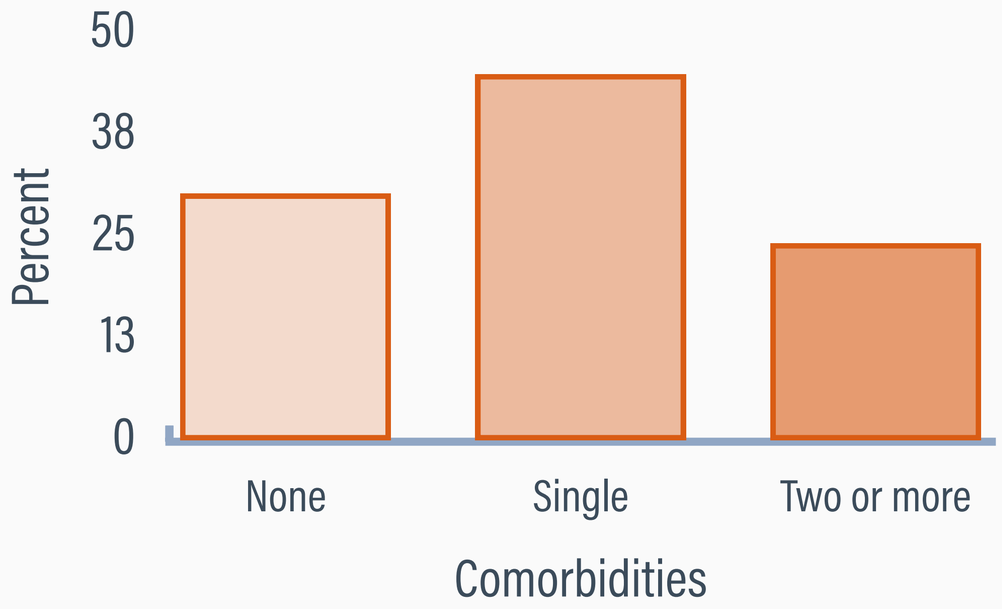
Score	Count	%	Cum. %
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Bar height (raw counts)

BAR GRAPH VS. FREQUENCY DISTRIBUTION

- Frequency distributions provide the same information in tabular format

Bar Graph



Frequency Distribution

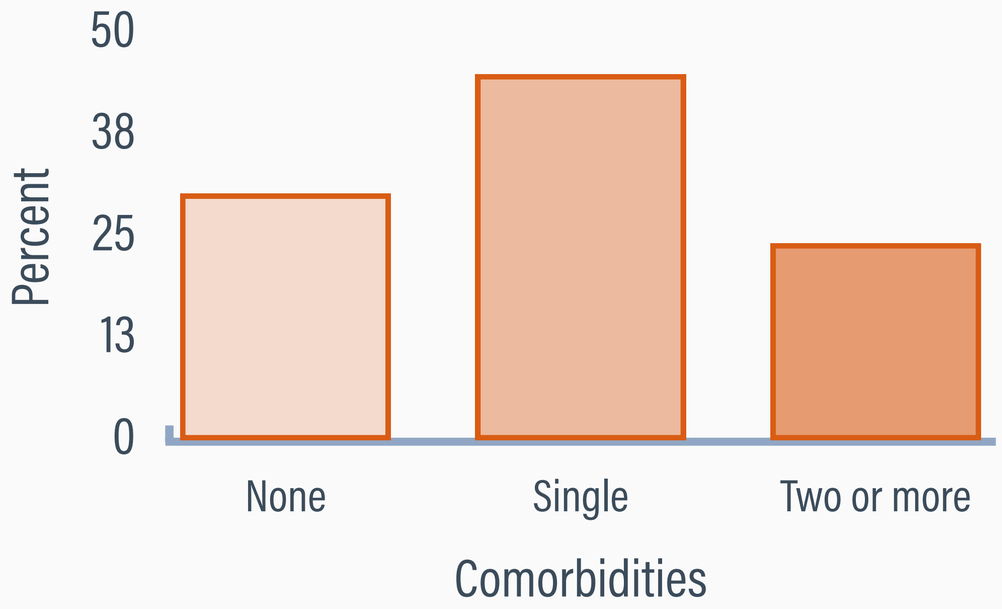
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Bar height (as %)

BAR GRAPH VS. FREQUENCY DISTRIBUTION

- Frequency distributions provide the same information in tabular format

Bar Graph



Frequency Distribution

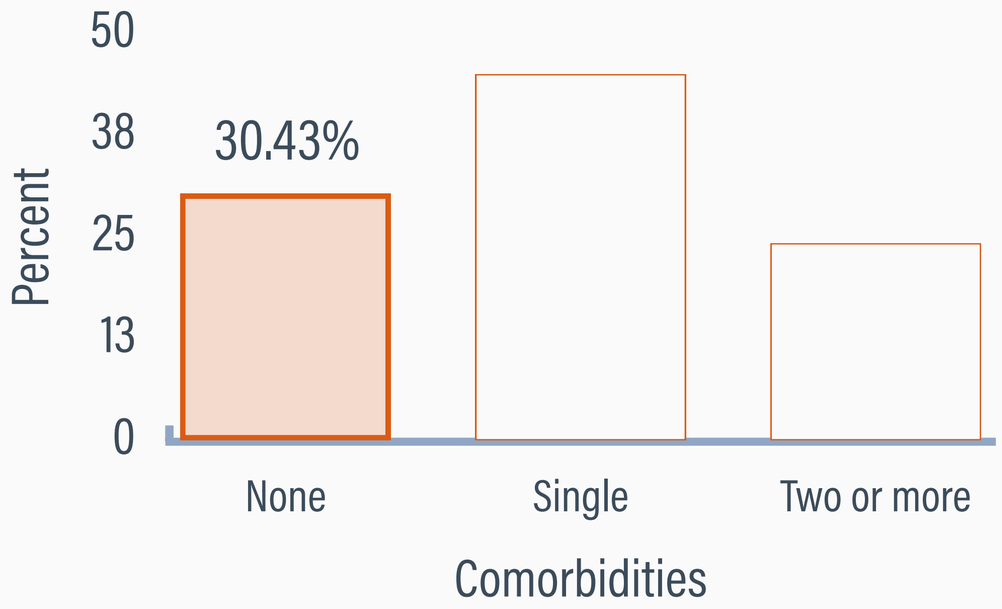
Score	Count	%	Cum. %
0 = None	91	30.43	30.43
1 = Single comorbidity	135	45.15	75.59
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Accumulate %s

CUMULATIVE (ACCUMULATING) PERCENTAGES

- Frequency distributions provide the same information in tabular format

Bar Graph



Frequency Distribution

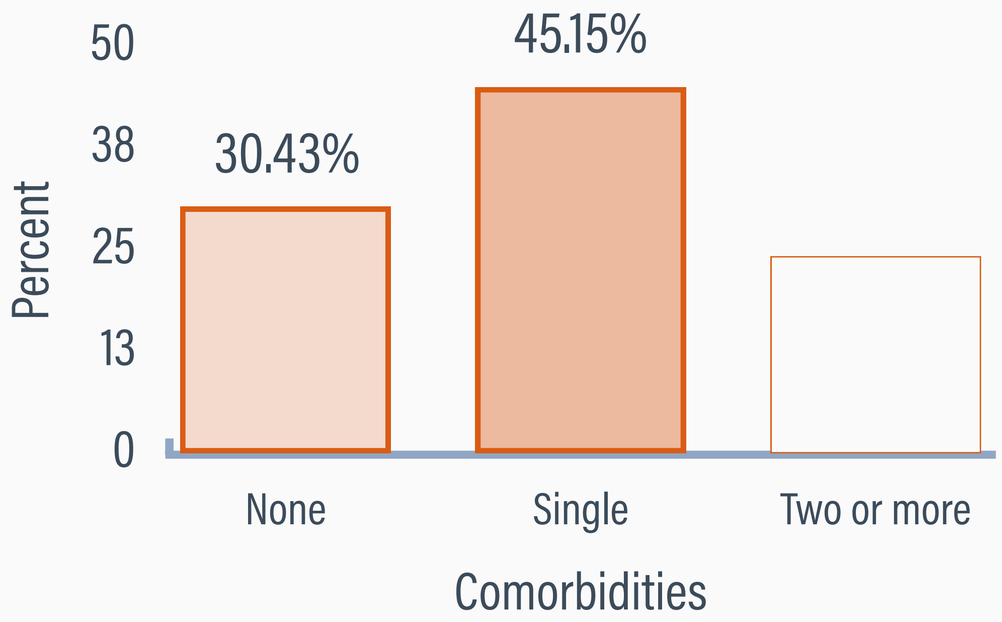
Score	Count	%	Cum. %
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Accumulate %s

CUMULATIVE (ACCUMULATING) PERCENTAGES

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Bar Graph



Frequency Distribution

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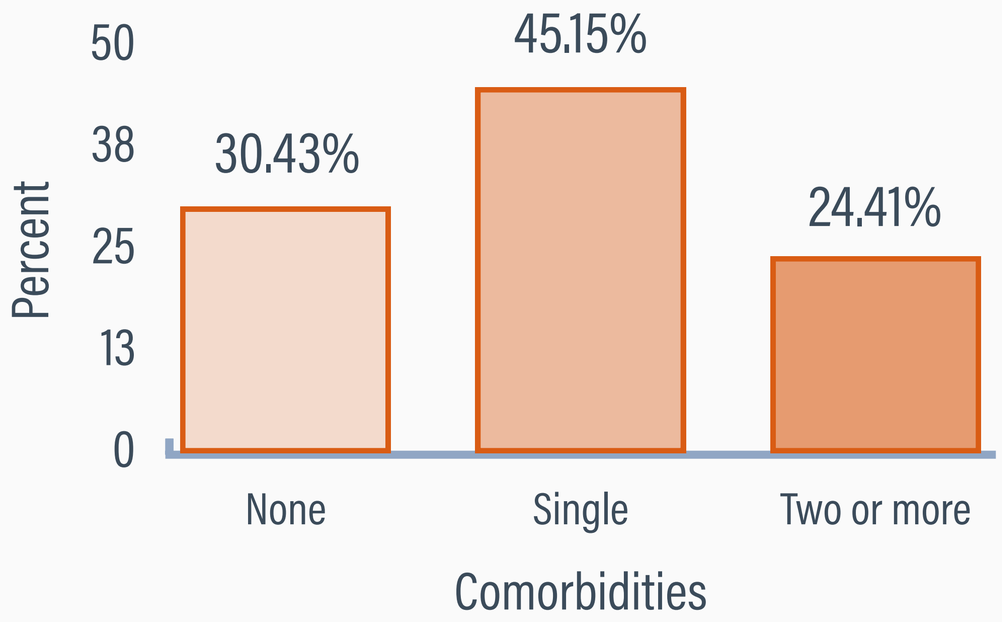
+

Accumulate %s

CUMULATIVE (ACCUMULATING) PERCENTAGES

- Frequency distributions provide the same information in tabular format

Bar Graph



Frequency Distribution

Score	Count		%	Cum. %
0 = None	91		**30.43**	30.43
1 = Single comorbidity	135	+	**45.15**	75.59
2 = Two or more	73	+	**24.41**	**100.0**

Accumulate %s

R OUTPUT

Frequencies
Cancer$Comorbids
Type: Integer

	Freq	% Valid	% Valid Cum.	% Total	% Total Cum.
0	91	30.43	30.43	30.43	30.43
1	135	45.15	75.59	45.15	75.59
2	73	24.41	100.00	24.41	100.00
<NA>	0			0.00	100.00
Total	299	100.00	100.00	100.00	100.00

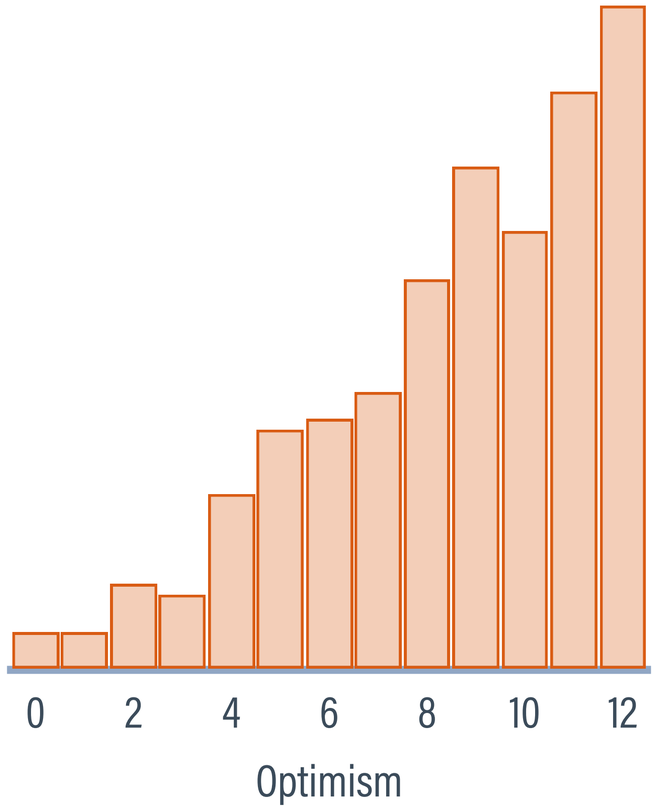


In small groups of two or three, discuss the optimism frequency distribution. What shape best describes the distribution? What features of the table convey the variable's shape?

Score	Count	%	Cum. %
0	3	1.0	1.0
1	3	1.0	2.0
2	7	2.3	4.4
3	6	2.0	6.4
4	14	4.7	11.0
5	19	6.4	17.4
6	20	6.7	24.1
7	22	7.4	31.4
8	31	10.4	41.8
9	40	13.4	55.2
10	35	11.7	66.9
11	46	15.4	82.3
12	53	17.7	100.0



Score	Count	%	Cum. %
0	3	1.0	1.0
1	3	1.0	2.0
2	7	2.3	4.4
3	6	2.0	6.4
4	14	4.7	11.0
5	19	6.4	17.4
6	20	6.7	24.1
7	22	7.4	31.4
8	31	10.4	41.8
9	40	13.4	55.2
10	35	11.7	66.9
11	46	15.4	82.3
12	53	17.7	100.0



In small groups of two or three, examine the frequency distribution and corresponding graph of the optimism scores. Evaluate the strengths and limitations of each representation for understanding the data. In your view, which provides more meaningful insight for summarizing this variable. Why?

OUTLINE

1. Measuring psychological constructs
2. Distribution shapes for numeric variables
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STUDY QUESTIONS

Mid-term and Finals would be from here!

- The study questions continue using the cancer research scenario introduced in the lecture, including new variables
- The questions illustrate the kinds of statistical reasoning the exams will require, including interpreting evidence, evaluating conclusions, and connecting graphical and numerical results
- Midterm exam questions will require you to apply these same reasoning skills to new research scenarios

STUDY QUESTIONS (1)

1. Suppose the researchers asked participants to rate their overall health using five categories ranging from 1 = poor to 5 = excellent. Sketch a histogram depicting a sample with relatively poor overall health.

2. Suppose instead that most participants reported very good or excellent health. What distribution shape would be consistent with this pattern? Provide a hypothetical skewness value consistent with that shape.

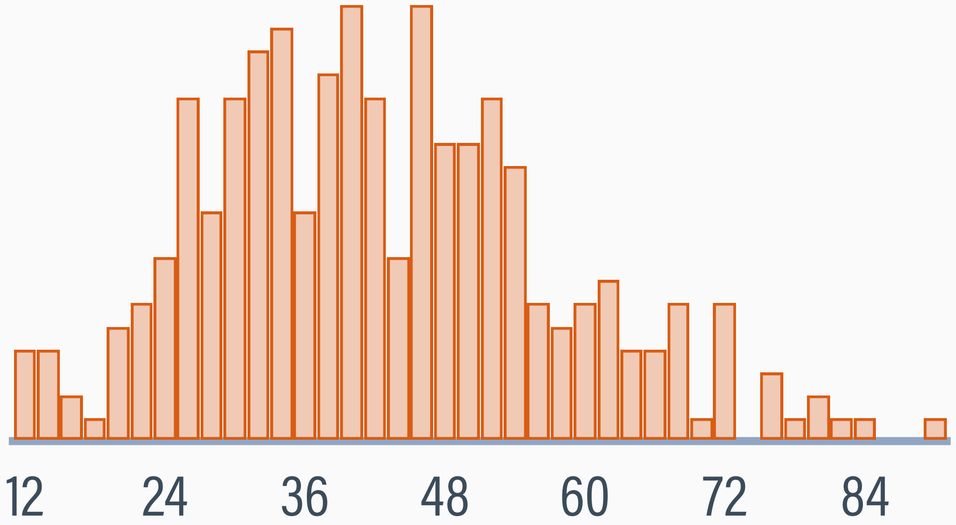
STUDY QUESTIONS (2)

3. Suppose the researchers measured quality of life using a sum score formed from several questionnaire items. Scores range from 0 to 30, with higher scores indicating better quality of life. Sketch a kernel density curve depicting a sample with relatively high quality of life.

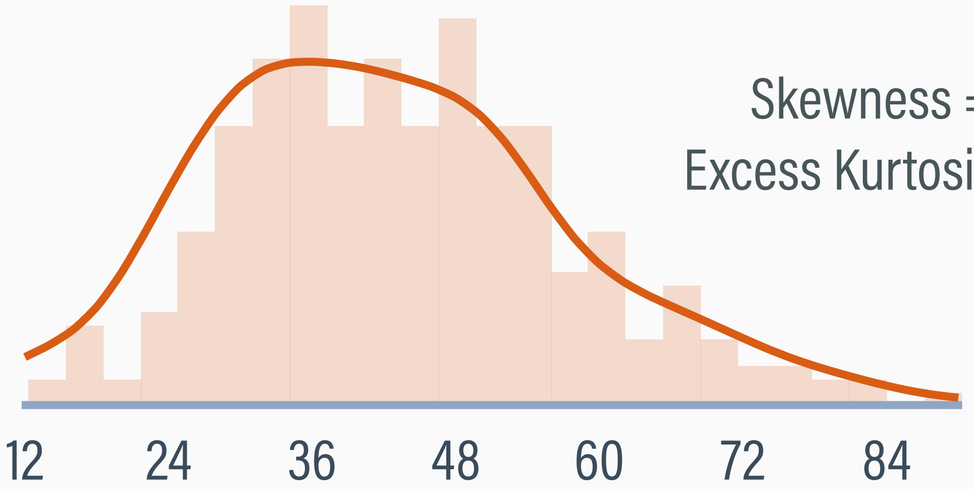
4. Suppose instead that most participants reported relatively poor quality of life. What distribution shape would be consistent with this pattern? Provide a hypothetical skewness value consistent with that shape.

VISUAL IMPAIRMENT DISTRIBUTION

Histogram



Kernel Density



STUDY QUESTIONS (3)

5. Based on the histogram and kernel density curve, how would you characterize the shape of the visual-impairment distribution? Explain what is meant by the “long tail” of a distribution and how it relates to distribution shape. Identify the specific graphical features that support your interpretation.

6. The visual-impairment distribution has a skewness of 0.44 and an excess kurtosis of –0.02. Interpret each statistic in practical terms. Explain how each value corresponds to the features shown in the histogram and kernel density curve.

VISUAL IMPAIRMENT FREQUENCY DISTRIBUTION

Frequencies
Cancer$VisImpair_grouped

	Freq	% Valid	% Valid Cum.	% Total	% Total Cum.
10-19	12	4.01	4.01	4.01	4.01
20-29	50	16.72	20.74	16.72	20.74
30-39	80	26.76	47.49	26.76	47.49
40-49	69	23.08	70.57	23.08	70.57
50-59	48	16.05	86.62	16.05	86.62
60-69	24	8.03	94.65	8.03	94.65
70-79	11	3.68	98.33	3.68	98.33
80-89	4	1.34	99.67	1.34	99.67
90-99	1	0.33	100.00	0.33	100.00
Total	299	100.00	100.00	100.00	100.00

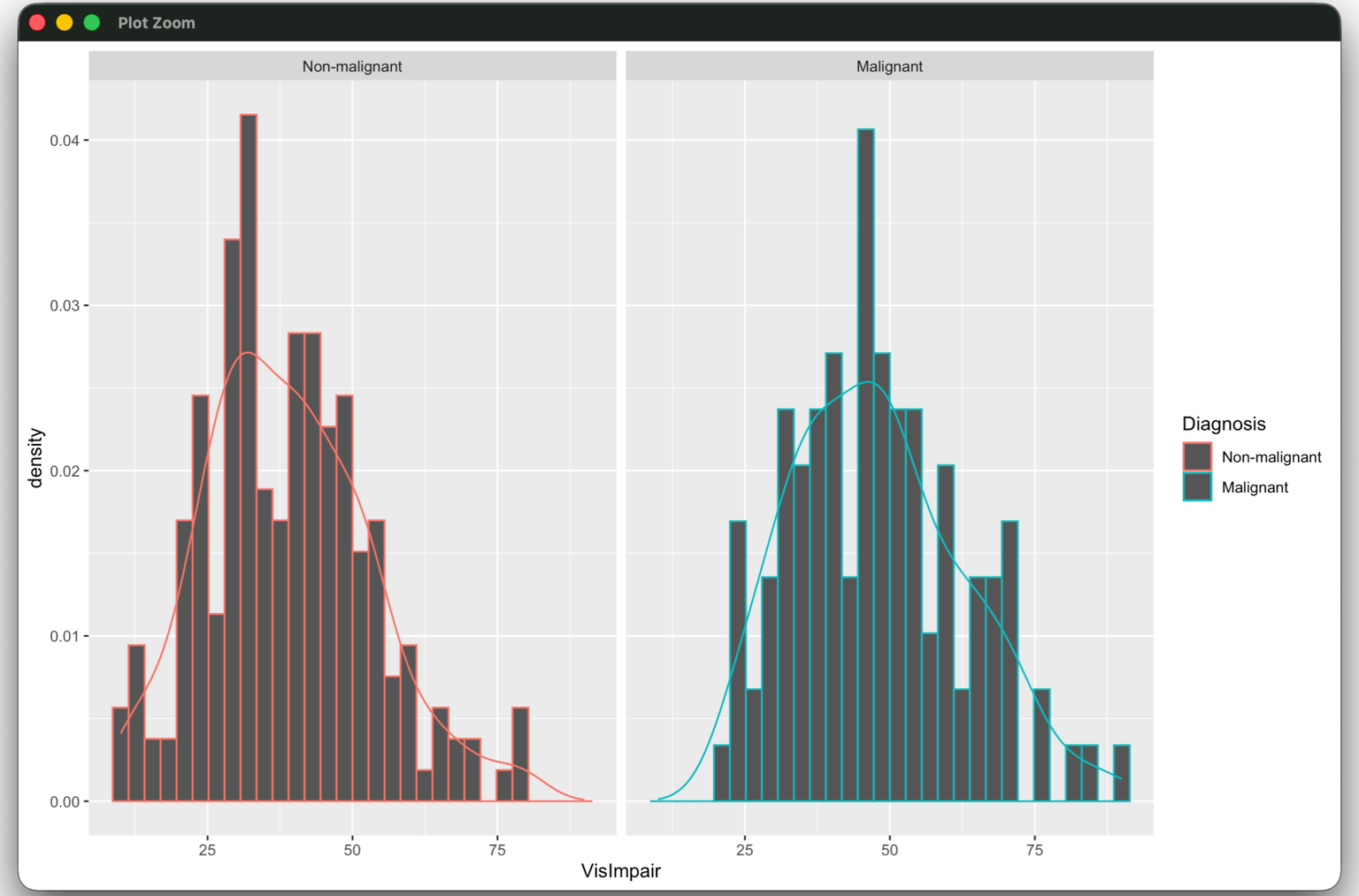
STUDY QUESTIONS (4)

7. In the grouped frequency distribution, the cumulative percentage for the 30–39 interval is 47.49%. Describe how this value is calculated. Explain what a cumulative percentage represents in a grouped frequency distribution, and interpret 47.49% in the context of this study.

8. In the same row, the valid percentage for the 30–39 interval is 26.76%. Describe how this value is calculated. Explain what a valid percentage represents, and interpret 26.76% in the context of this study.

9. How do the frequencies or percentages across the score intervals reflect the shape shown in the histogram and density curve? Identify specific patterns in the grouped frequency distribution that support the same conclusion about the shape of the visual-impairment distribution.

VISUAL IMPAIRMENT BY DIAGNOSIS



STUDY QUESTIONS (5)

10. Suppose researchers want to determine whether patients with malignant and non-malignant diagnoses differ in visual impairment. State your hypothesis: Would you expect visual impairment to be higher (worse) in the malignant group, higher in the non-malignant group, or similar across the two groups? Briefly explain your prediction.

11. Is your hypothesis consistent or inconsistent with the side-by-side histograms and density curves? Support your conclusion by comparing the distributions' location, spread, shape, and overlap.